

Descent of étale morphisms. Application to the fundamental group

SGA 1, Exposé IX

English draft

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About this draft

This is an unofficial English translation of Exposé IX, “Descente des morphismes étales. Application au groupe fondamental”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ IX

Descent of étale morphisms. Application to the fundamental group

1. Recollections on étale morphisms

We shall review here the properties of étale morphisms (developed in I) that will be of use to us, taking this opportunity to eliminate superfluous noetherian hypotheses from the theory. The reader will note that even if one is interested only in noetherian schemes, the technique of descent leads one to introduce non-noetherian schemes (such as $\mathrm{Spec}(\hat{A} \otimes_A \hat{A})$, where A is a noetherian local ring), and in order to be able to apply the language of fibered categories, it is important to define notions such as étale morphism etc., without introducing any noetherian restriction into them. The reader who would be reluctant to verify or to accept that the statements below are true without noetherian hypotheses may content himself with accepting them under the noetherian hypotheses of I, provided he introduces these same noetherian hypotheses into the statements of the following nos., and uses Definition 1.1 below for the non-noetherian schemes that arise in the arguments.

DEFINITION 1.1. Let $f: X \rightarrow S$ be a morphism of preschemes, and x a point of X . We say that f is *étale at x* , or that X is *étale over S at x* , if there exist an affine open neighborhood U of $s = f(x)$, an affine open neighborhood V of x over U , a *noetherian* affine scheme U_0 , an *étale* (I) and affine U_0 -scheme V_0 , a morphism $U \rightarrow U_0$, and a U -isomorphism

$$V \xrightarrow{\sim} V_0 \times_{U_0} U.$$

Note that when S is locally noetherian, this terminology coincides with that of loc. cit. We shall likewise say that f is *étale*, or that X is *étale over S* , if f is étale at every point x of X . With these definitions, the propositions below reduce without difficulty to the noetherian case, where they are proved in I no. 4, 5, 7. For details, the reader may consult EGA IV¹.

REMARKS 1.2. If f is étale at x , then f is “*of finite presentation at x* ” (VIII 3.5), the local ring of x in the fiber $f^{-1}(s)$ is a *finite separable extension* of $k(s)$, and finally f is *flat at x* . One can show that the converse is true, hence that Definition 1.1 is the same as in the case where S is locally noetherian, except that the condition “of finite type at x ” must be replaced by “of finite presentation at x ”. Since this result has a delicate proof, we did not wish to give here this definition of the notion of étale morphism, which does not lend itself directly to the proof of the properties that will follow.

Let us first note that we trivially have:

PROPOSITION 1.3. *If $f: X \rightarrow S$ is étale, then every morphism $f': X' \rightarrow S'$ deduced from it by a change of base $S' \rightarrow S$ is also étale.*

We may therefore say that the étale morphisms form a *fibered subcategory* of the category of arrows in **Sch** (cf. VI 11 a)). The object of the present exposé is the study of the exactness properties of this fibered category over **Sch**.

PROPOSITION 1.4. *Let $f: X \rightarrow S$ be a morphism of preschemes. For it to be an open immersion, it is necessary and sufficient that it be étale and radicial.*

¹More precisely, EGA IV 17, 18.

Cf. I 5.1. We conclude from this that if X is étale over S , every section of X over S is an open immersion, hence, using 1.4 again, we find:

COROLLARY 1.5. *Let X be an étale S -prescheme. Then there is a one-to-one correspondence between the set of sections of X over S , and the set of open subsets Γ of X such that the morphism $\Gamma \rightarrow S$ induced by the structural morphism is radicial and surjective.*

If moreover X is separated over S , Γ will be a subset of X that is both open and closed, but this does not matter. — Making an obvious change of base, one can put 1.5 in the apparently more general form:

COROLLARY 1.6. *Let X and Y be two S -preschemes, Y being étale over S . Then the map $f \mapsto \Gamma_f$ which associates to every S -morphism f from X to Y the subset of $X \times_S Y$ underlying the graph of f , is a bijection of $\text{Hom}_S(X, Y)$ onto the set of open subsets of Γ of $X \times_S Y$ such that the morphism $\Gamma \rightarrow X$ induced by pr_1 is radicial and surjective.*

PROPOSITION 1.7. *Let S_0 be the subprescheme of S defined by a quasi-coherent Nil-ideal, i.e. such that S_0 has the same underlying set as S . Then the functor $X \mapsto X \times_S S_0$ from the category of preschemes étale over S to the category of preschemes étale over S_0 is an equivalence of categories.*

The fact that this functor is fully faithful is an immediate consequence of 1.6. The fact that it is essentially surjective is contained in I 8.3. Note that in the preceding equivalence, X is of finite type i.e. quasi-finite over S , (resp. finite i.e. an étale covering of S), if and only if X_0 satisfies the analogous condition over S_0 ; same remark for the separation condition. These facts are immediate, and are also contained in 2.4 below.

COROLLARY 1.8. *Let A be a complete noetherian local ring with residue field k . Then the functor $B \mapsto B \otimes_A k$ is an equivalence of the category of algebras finite and étale over A with the category of algebras finite and étale over k , (i.e. composed of a finite number of finite separable extensions of k).*

PROPOSITION 1.9. *For X to be an étale covering of S i.e. finite and étale over S , it is necessary and sufficient that X be S -isomorphic to the spectrum of an Algebra $\mathcal{A}$ over S which is a locally free Module of finite type, and such that for every $s \in S$, $\mathcal{A}_s \otimes_{\mathcal{O}_s} k(s)$ is a separable algebra over $k(s)$, hence in the present case a direct product of finite separable extensions of $k(s)$.*

Finally the following result is of a less elementary nature, being the conjunction of I 8.4 and of the *existence theorem for sheaves in algebraic geometry*, (EGA III 5; cf. also [1] th. 3).

THEOREM 1.10. *Let S be the spectrum of a complete noetherian local ring, X a proper S -scheme, X_0 the fiber of X at the closed point of S , (so that X_0 is a closed subscheme of X). Then the restriction functor $X' \mapsto X' \times_X X_0$ is an equivalence of the category of étale coverings of X with the category of étale coverings of X_0 .*

2. Submersive and universally submersive morphisms

DEFINITION 2.1. A morphism $g: S' \rightarrow S$ of preschemes is said to be *submersive* if it is surjective, and makes S a topological quotient space of S' (i.e. a subset U of S such that $f^{-1}(U)$ is open, is open). We say that f is *universally submersive* if for every morphism $T \rightarrow S$, the morphism $f': T' = S' \times_S T \rightarrow T$ deduced from f by change of base is submersive.

It is immediate that the composite of two submersive morphisms (resp. universally submersive morphisms) is submersive (resp. universally submersive), and that a change of base in a universally submersive morphism gives a universally submersive morphism (since one has done what is needed for this). If fg is submersive (resp. universally submersive), f is.

EXAMPLES 2.2. a) A surjective morphism which is open, or closed, is submersive, hence a surjective universally closed or universally open morphism is universally submersive. For example a *surjective proper morphism is universally submersive*. On the other hand a *faithfully flat and quasi-compact morphism is universally submersive* (VIII 4.3). These will be the two most important cases for us.

One can apply to a submersive or universally submersive morphism $g: S' \rightarrow S$ the arguments of VIII 4.3; one finds in particular:

PROPOSITION 2.3. *Suppose $g: S' \rightarrow S$ submersive. Then the following diagram of maps is exact:*

$$\mathrm{Ouv}(S) \longrightarrow \mathrm{Ouv}(S') \rightrightarrows \mathrm{Ouv}(S''),$$

where $S'' = S' \times_S S'$, and where $\mathrm{Ouv}(X)$ denotes the set of open subsets of the prescheme X .

PROPOSITION 2.4. *Let $g: S' \rightarrow S$ be a universally submersive morphism, $f: X \rightarrow Y$ an S -morphism, and $f': X' \rightarrow Y'$ the S' -morphism deduced from it by change of base. For f to be open (resp. closed), it suffices that f' be so. For f to be universally open, resp. universally closed, resp. separated, it is necessary and sufficient that f' be so. If moreover g is quasi-compact, and f locally of finite type, for f to be proper it is necessary and sufficient that f' be so.*

For this last point, one notes that if f' is proper, hence quasi-compact, then f is quasi-compact (VIII 3.3), hence of finite type since it is locally of finite type. On the other hand it is separated and universally closed by what precedes, hence it is proper.

PROPOSITION 2.5. *Let S' be a prescheme of finite type over the spectrum S of a complete noetherian local ring, suppose that the fiber of the closed point s of S is finite, hence the local rings in S' of the points s' of this fiber are finite over $A = \mathcal{O}_s$. Let S'' be the scheme sum of the spectra of the $\mathcal{O}_{S',s'}$ in question, considered as a finite S -scheme. For $g: S' \rightarrow S$ to be universally submersive, it is necessary and sufficient that the structural morphism $S'' \rightarrow S'$ be surjective.*

Since there is a natural S -morphism $S'' \rightarrow S'$, and a finite surjective morphism is universally submersive by 2.2, the stated condition is sufficient. Let us therefore show that if $S'' \rightarrow S'$ is not surjective, then g is not universally submersive. Indeed, let t be a point of S which is not in the image of S'' ; there then exists an S -scheme T , spectrum of a discrete valuation ring, whose image in S is $\{s, t\}$. Note that the image of S'' in S' is open, since the morphism $S'' \rightarrow S'$ is a local isomorphism, and on the other hand this image contains S'_s , and does not meet S'_t . It follows that the inverse image of the latter in $T' = S' \times_S T$ is *open*, and identical to the inverse image of the closed point of T . This shows that $T' \rightarrow T$ is not submersive, hence $S' \rightarrow S$ is not universally submersive.

REMARK 2.6. Using the criterion IV 6.3 for a constructible subset of a noetherian space to be open, one easily finds the following valuative criterion for a morphism $g: S' \rightarrow S$ of *finite type*, with S locally noetherian, to be universally submersive: it is necessary and sufficient that for every S -scheme T , spectrum of a discrete valuation ring, setting $T' = S' \times_S T$, the inverse image in T' of the closed point of T be non-open.

3. Descent of morphisms of étale preschemes

PROPOSITION 3.1. *Let $g: S' \rightarrow S$ be a surjective morphism of preschemes, X and Y two preschemes over S , X', Y' their inverse images over S' . If Y is unramified over S , then the canonical map*

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y')$$

is injective.

Indeed, by virtue of 1.6, an S -morphism $f: X \rightarrow Y$ is known when one knows the set underlying its graph Γ , which is a subset of $Z = X \times_S Y$. Since

$$Z' = Z \times_S S' = X' \times_{S'} Y' \longrightarrow Z$$

is surjective, (since $S' \rightarrow S$ is), this subset Γ is known when one knows its inverse image in $X' \times_{S'} Y'$, which is none other than the set underlying the graph of f' . Whence the conclusion.

A subset Γ of Z is the graph of an S -morphism $f: X \rightarrow Y$ if and only if it is open, and if the morphism induced by pr_1 from Γ to Y is radicial and surjective cf. 1. When the first property is satisfied, the second one is if and only if the inverse image Γ' of Γ in Z' satisfies the same condition VIII 3.1. If one knows finally that $Z' \rightarrow Z$ is submersive, which will be the case in particular if $S' \rightarrow S$ is universally submersive, then Γ is open if and only if Γ' is. Thus, the set $\text{Hom}_S(X, Y)$ is then in one-to-one correspondence with the set of open subsets Γ' of Z' such that the projection morphism $\text{pr}_1: Z' \rightarrow X'$ is radicial and surjective, (i.e. corresponding to an S' -morphism $f': X' \rightarrow Y'$), and which are saturated for the equivalence relation defined by $Z' \rightarrow Z$, i.e. whose two inverse images in $Z'' = Z' \times_Z Z' = Z \times_S S''$ (where $S'' = S' \times_S S'$), by the one and the other projection, are equal. Now these latter are the graphs of the two S'' -morphisms $X'' \rightarrow Y''$ deduced from f' by change of base, by the one and the other projection $S'' \rightarrow S'$. We have thus obtained:

PROPOSITION 3.2. *Let $g: S' \rightarrow S$ be a universally submersive morphism of preschemes, $S'' = S' \times_S S'$, X and Y two S -preschemes, X' and Y' their inverse images over S' , and X'' , Y'' their inverse images over S'' . If Y is étale over S , the following canonical diagram of maps is exact:*

$$\text{Hom}_S(X, Y) \longrightarrow \text{Hom}_{S'}(X', Y') \rightrightarrows \text{Hom}_{S''}(X'', Y'').$$

Taking X and Y étale over S , one finds the following statement, which moreover gives back 3.2 (even when restricting to $X = S$, a case to which one can indeed always reduce in 3.2, by the change of base $X \rightarrow S$);

COROLLARY 3.3. *A universally submersive morphism of preschemes is a descent morphism for the fibered category of preschemes étale over others.*

I do not know, moreover, whether it is necessarily an *effective* descent morphism for the fibered category in question, even when one makes in addition the hypothesis that S is noetherian and g of finite type, and restricts oneself to étale coverings. We shall nevertheless give in the following no. some useful criteria of effectiveness.

COROLLARY 3.4. *Let $g: S' \rightarrow S$ be a universally submersive morphism whose fibers $g^{-1}(s)$ are “geometrically connected”, i.e. for every extension $K/k(s)$, $g^{-1}(s) \otimes_{k(s)} K$ is connected. Then S' is connected if S is. The functor from the category of preschemes étale over S to the category of preschemes étale over S' defined by g is fully faithful.*

A subset of S' which is both open and closed is saturated for the set-theoretic equivalence relation defined by g , since the fibers are connected, hence is the inverse image of a subset of S , which is necessarily open and closed since g is submersive. If therefore S is connected, S' is. This also implies the following result: the composite fg of two morphisms with universally connected fibers, f being universally submersive, has universally connected fibers; if S'_1 and S'_2 over S have universally connected fibers, the same holds for $S'_1 \times_S S'_2$. In particular, under the conditions of 3.4, S'' has universally connected fibers over S . Let then X and Y be étale over S , and let u' be an S' -morphism from X' to Y' ; let us prove that it is compatible with the descent data (which implies the desired conclusion thanks to 3.3). Now let u''_1 and u''_2 be the two S'' -morphisms $X'' \rightarrow Y''$ deduced from u' . The subprescheme of S'' of coincidences of u''_1 and u''_2 is an induced open subprescheme, closed fiber by fiber, as the inverse image

of the diagonal prescheme of Y'' over S''^2 . It is therefore the inverse image of a subset of S . Since it contains the diagonal in S'' , it is identical to S'' , whence $u_1'' = u_2''$, qed.

4. Descent of étale preschemes: effectiveness criteria

PROPOSITION 4.1. *Let $g: S' \rightarrow S$ be a faithfully flat and quasi-compact morphism. Then g is an effective descent morphism for the fibered category of preschemes étale, separated and of finite type over others.*

It is indeed a descent morphism for the fibered category in question, by virtue of 3.3 or of VIII 5.2, as one prefers. It remains to show that if X' is étale, separated and of finite type over S' , and endowed with a descent datum relative to $g: S' \rightarrow S$, the latter is effective in the fibered category in question. Now one easily sees that if X is a prescheme over S , then it is étale over S if and only if it is étale over S' (by virtue of definition 1.1 and of loc. cit. 3.6). Hence it is étale, separated and of finite type over S if and only if X' is so over S' , cf. for example 2.4. Hence it suffices to make sure of the effectiveness of the descent datum on X for the fibered category of arrows of **Sch**. Now X' is quasi-affine over S' by virtue of VIII 6.2 and 6.6. One can then conclude by using VIII 7.9. The reader will note moreover that the proof requires less if one restricts oneself to preschemes étale and *finite* over others, for one can then invoke VIII 2.1 directly.

COROLLARY 4.2. *Let $g: S' \rightarrow S$ be a universally submersive morphism, X' an S' -prescheme étale separated and of finite type, endowed with a descent datum relative to g , $S_1 \rightarrow S$ a faithfully flat and quasi-compact morphism, S'_1 and X'_1 deduced from S' and X' by change of base, so that $S'_1 \rightarrow S_1$ is universally submersive, X'_1 is étale separated and of finite type over S'_1 , and endowed with a descent datum relative to $g_1: S'_1 \rightarrow S_1$. For the descent datum on X' to be effective, it is necessary and sufficient that the descent datum on X'_1 be so.*

This follows from the theory of descent in categories **[D]**, taking into account 4.1 and 3.3. One proves in an analogous way:

COROLLARY 4.3. *Let $g: S' \rightarrow S$ be a universally submersive morphism, X' an étale S' -prescheme endowed with a descent datum relative to g , (S_i) a covering of S by open sets. For the descent datum to be effective, it is necessary and sufficient that for every i , the corresponding descent datum on $X'_i = X' \times_S S_i$, relative to the morphism $g_i: S'_i = S' \times_S S_i \rightarrow S_i$, be so.*

This last result leads to bringing out a local effectiveness criterion:

PROPOSITION 4.4. *Let $g: S' \rightarrow S$ be a morphism of finite presentation VIII 3.6 and universally submersive, X' a prescheme étale and of finite presentation over S' , endowed with a descent datum relative to g , and finally a a point of S . For there to exist an open neighborhood U of a such that the corresponding descent datum on $X'_U = X' \times_S U$ relative to the morphism*

$$g_U: S'_U = S' \times_S U \longrightarrow S_U = U$$

is effective, it is necessary and sufficient that the corresponding descent datum on $X'_a = X' \times_S \text{Spec}(\mathcal{O}_a)$, relative to the morphism

$$g_a: S'_a = S' \times_S \text{Spec}(\mathcal{O}_a) \longrightarrow S_a = \text{Spec}(\mathcal{O}_a),$$

be effective.

The necessity being trivial, let us show the sufficiency. We therefore have an étale prescheme of finite type X_a over S_a , and an isomorphism

$$(*) \quad X'_a \xrightarrow{\sim} X_a \times_{S_a} S'_a$$

²note that the fibers of S' over S are separated!

compatible with the descent data. In accordance with an easy general sorites on preschemes defined over an inductive limit of rings (here the rings A_f , where A is the ring of an affine open neighborhood of a , and where f runs through the elements of A that are not in the prime ideal corresponding to a), one can find an open neighborhood U of a , an étale prescheme of finite type X_U over $U = S_U$, and an S_a -isomorphism $X_a \xrightarrow{\sim} X_U \times_{S_U} S_a$. Moreover, taking U small enough, one can then assume that the isomorphism $(*)$ comes from an isomorphism:

$$X'_U \xrightarrow{\sim} X_U \times_{S_U} S'_U;$$

the latter might not be compatible with the descent data; however, provided one shrinks U , it will be compatible with the descent data. This completes the proof.

COROLLARY 4.5. *Under the conditions of 4.4, for the descent datum on X' to be effective, it is necessary and sufficient that for every $a \in S$, the corresponding descent datum on X'_a , relative to the morphism $S'_a = S' \times_S \text{Spec}(\mathcal{O}_a) \rightarrow \text{Spec}(\mathcal{O}_a)$, be so. When S is locally noetherian, and X' separated over S' , one can in the preceding criterion also replace $\mathcal{O}_a$ by its completion.*

The first assertion follows from 4.4 and 4.3, the second is then a consequence of 4.2. Using 4.2 again and the fact that for every noetherian local ring A , one can find a complete noetherian local ring B and a local homomorphism $A \rightarrow B$ such that B is flat over A and $B/\mathfrak{m}B$ is a given extension of the residue field $k = A/\mathfrak{m}$ of A , one finds:

COROLLARY 4.6. *Under the conditions of 4.4, assume moreover X' separated over S' , and S locally noetherian. For the descent datum on X' to be effective, it is necessary and sufficient that for every prescheme S_1 over S which is the spectrum of a complete local ring with algebraically closed residue field, the corresponding descent datum on $X'_1 = X' \times_S S_1$, relative to the morphism $g_1: S'_1 \rightarrow S_1$, be effective.*

THEOREM 4.7. *Let $g: S' \rightarrow S$ be a morphism which is finite and surjective, and of finite presentation (this last hypothesis being a consequence of the others if S is locally noetherian)³. Then g is an effective descent morphism for the fibered category of preschemes étale, separated, of finite type over others.*

One must show that if X' is étale, separated, of finite type over S' , and endowed with a descent datum relative to g , then this datum is effective. Using 4.3, one easily reduces to the case where S is noetherian. Thanks to 4.5, one can therefore assume that S is the spectrum of a noetherian local ring, a fortiori that

$$\dim S = n < +\infty.$$

One then argues by induction on $\dim S = n$, the assertion being trivial for $n < 0$. Assume therefore $n \geq 0$ and the theorem proved for the dimensions $n' < n$. By virtue of 4.6 one is reduced to the case where S is the spectrum of a complete local ring, hence S' is a finite union of spectra of complete local rings. We therefore have

$$X' = X'_1 \sqcup X'_2$$

where X'_1 is *finite* over S' , and where X'_2 has no point lying over any of the closed points of S' . Consider the morphisms

$$q_1, q_2: X'' \rightrightarrows X'$$

corresponding to the descent datum, compatible with $p_1, p_2: S'' \rightrightarrows S'$. One sees at once that

$$X'' = q_i^{-1}(X'_1) \sqcup q_i^{-1}(X'_2) \quad i = 1, 2$$

is the analogous canonical decomposition of X'' over S'' , which implies $q_1^{-1}(X'_1) = q_2^{-1}(X'_1)$ and consequently X'_1 and X'_2 are endowed with induced descent data. Now let T be the

³One can show that it in fact suffices that g be an *integral* morphism, by reducing to the case of the text by a limit procedure in the style of EGA IV 8.

open set of S complementary to its closed point, so that $T' = S' \times_S T$ is the part of S' complementary to the set of closed points, and X'_2 , which lies entirely over T' , is endowed with a descent datum relative to the morphism $T' \rightarrow T$ induced by g . As the latter is finite surjective, and as $\dim T < \dim S = n$, this descent datum is effective by the induction hypothesis. One therefore sees that it suffices to prove that the descent datum on X'_1 is effective, hence one can now assume X' étale and *finite* over S' . (N.B. the argument by induction is unnecessary if one restricts oneself to étale coverings in statement 4.7). Let then S_0 be the spectrum of the residue field of A , let $S'_0 = S' \times_S S_0$, and define similarly S''_0, S'''_0 starting from the fibered squares and cubes S'' and S''' of S' over S . By virtue of 1.8, the morphisms $S_0 \rightarrow S$, $S'_0 \rightarrow S'$, etc. induce equivalences for the categories of étale coverings of S and S_0 on the one hand, S' and S'_0 on the other hand, etc. By the sorites of the theory of descent in categories [D], it follows that for $g: S' \rightarrow S$ to be an effective descent morphism for the fibered category of étale coverings, it is necessary and sufficient that the same hold for $g_0: S'_0 \rightarrow S_0$. But this is indeed the case, as a special case of 4.1, for example. This completes the proof.

COROLLARY 4.8. *The conclusion of 4.7 remains valid if one only assumes that $S' \rightarrow S$ is universally submersive, of finite type and quasi-finite, provided one assumes S locally noetherian.*

By virtue of 4.6, one can indeed assume that S is the spectrum of a complete noetherian local ring. Then by virtue of 2.5, there exist a finite and surjective morphism $S_1 \rightarrow S$ and an S -morphism $S_1 \rightarrow S'$. As $S_1 \rightarrow S$ is a universal strict descent morphism for the fibered category under consideration, by virtue of 4.7, and as $S' \rightarrow S$ is a universal descent morphism for the said category, 4.8 follows from the general sorites [D].

COROLLARY 4.9. *Let $g: S' \rightarrow S$ be a morphism of finite type, surjective and universally open, with S locally noetherian. Then g is an effective descent morphism for the fibered category of preschemes étale, separated and of finite type over others.*

Proceeding as in 4.7, one is reduced to the case where S is the spectrum of a noetherian and complete local ring A . Let A_1 be a finite algebra over A , with spectrum S_1 , such that $S_1 \rightarrow S$ is finite and *surjective*, hence a universal effective descent morphism for the fibered category under consideration, thanks to 4.7. It then follows from the general theorems [D] that g is an effective descent morphism for the fibered category under consideration if and only if the corresponding morphism $g_1: S'_1 = S' \times_S S_1 \rightarrow S_1$ is so. As the latter satisfies the same hypotheses as g , one is reduced to proving 4.9 for S_1 instead of S . Taking first for A_1 the direct product of the $A/\mathfrak{p}_i$, for the minimal prime ideals $\mathfrak{p}_i$ of A , one is reduced to the case where A is *integral*. One then shows⁴ that there exists an integral subscheme S_1 of S' , quasi-finite over S and dominating S , passing through a point of the fiber of S' at the closed point y of S (thanks to the fact that S' is universally open of finite type over S local noetherian integral, and $S'_y \neq \emptyset$). As A is complete, S_1 is finite over S , and as it dominates S , the morphism $S_1 \rightarrow S$ is surjective. Replacing S by S_1 once more, one is reduced to the case where S' has a section over S , where the statement is trivial.

THEOREM 4.10. *Let $g: S' \rightarrow S$ be a finite radicial surjective morphism, of finite presentation (this last condition being superfluous if S is locally noetherian⁵). Then the inverse image functor induces an equivalence of the category of étale preschemes over S with the category of étale preschemes over S' .*

As the diagonal morphisms from S' into $S' \times_S S'$ and $S' \times_S S' \times_S S'$ are surjective immersions, hence induce, by virtue of 1.9, equivalences of the categories of étale preschemes over $S' \times_S S'$ resp. $S' \times_S S' \times_S S'$ with the category of étale preschemes over S' , it follows

⁴Cf. EGA IV 14.3.13 and 14.5.4.

⁵It even suffices that g be integral, radicial surjective, as one sees by an easy reduction to the case of the text, in the style of EGA IV 8, cf. SGA 4 VIII 1.1.

from the sorites of descent [D] that every X' étale over S' is endowed with one and only one descent datum relative to $g: S' \rightarrow S$. Hence 3.3 implies that the inverse image functor by g , from the category of étale preschemes over S into the category of étale preschemes over S' , is *fully faithful*. It remains to show that it is essentially surjective, i.e. that every X' étale over S' is isomorphic to the inverse image of an X étale over S . The question being evidently local on S and on X' , one can assume S, S', X' affine. But then X' is separated of finite type over S' , and one can apply the effectiveness criterion 4.7.

COROLLARY 4.11. *The conclusion 4.9 remains valid when one replaces the hypothesis on g by: g is faithfully flat, quasi-compact and radicial.*

Same proof, invoking 4.1 instead of 4.7.

One will note that the proof of 4.7 is “elementary” in that it does not use the finiteness and comparison theorems for proper morphisms (EGA III 3, 4, 5). The same no longer holds for the following result:

THEOREM 4.12. *Let $g: S' \rightarrow S$ be a proper, surjective morphism, of finite presentation (this last hypothesis being a consequence of the first if S is locally noetherian). Then g is an effective descent morphism for the fibered category of étale coverings of preschemes.*

By virtue of 3.3 and of 2.2, one is reduced to proving that for every étale covering X' over S' , endowed with a descent datum relative to $g: S' \rightarrow S$, this descent datum is effective. Using 4.3, one is easily reduced to the case where S is noetherian, and using 4.6, one can therefore assume that S is the spectrum of a *complete* noetherian local ring A . Let us introduce S'' and S''' as usual, let S_0 be the spectrum of the residue field of A , and let S'_0, S''_0, S'''_0 be deduced from S', S'', S''' by the change of base $S_0 \rightarrow S$, i.e. the fibers of S', S'', S''' at the closed point of S . By 1.10, the morphisms $S_0 \rightarrow S, S'_0 \rightarrow S'$, etc. induce equivalences of the category of étale coverings over the target scheme with the category of étale coverings over the source scheme. Consequently, $g: S' \rightarrow S$ is a strict descent morphism for the fibered category of étale coverings of preschemes if and only if $g_0: S'_0 \rightarrow S_0$ is, which is indeed the case by virtue of 4.1. This completes the proof of 4.12. (In this argument, we needed from 1.10 only the fact that the functor considered in 1.10 is *fully faithful*, which does *not* use the existence theorem for coherent sheaves in algebraic geometry.)

5. Translation in terms of the fundamental group

Let

$$g: S' \longrightarrow S$$

be an *effective descent morphism* for the fibered category of *étale coverings* of preschemes, for example a proper, surjective morphism of finite presentation (4.12), or a faithfully flat and quasi-compact morphism. Introducing as usual S'', S''' , and denoting by $\mathcal{C}, \mathcal{C}', \mathcal{C}'', \mathcal{C}'''$ the category of étale coverings of S, S', S'', S''' respectively, we therefore have a 2-exact diagram of categories

$$(*) \quad \mathcal{C} \xrightarrow{p^*} \mathcal{C}' \xrightarrow{p_1^*, p_2^*} \mathcal{C}'' \xrightarrow{p_{21}^*, p_{32}^*, p_{31}^*} \mathcal{C}'''$$

corresponding to the diagram

$$S \xleftarrow{p} S' \xleftarrow{p_1, p_2} S'' \xleftarrow{p_{21}, p_{32}, p_{31}} S'''.$$

Suppose the preschemes S, S', S'', S''' are disjoint sums of connected preschemes, which will be the case in particular if they are locally connected preschemes, a fortiori if they are locally noetherian (for example if S' is of finite type over S locally noetherian). Then the categories $\mathcal{C}, \mathcal{C}' \dots$ in $(*)$ are multi-Galois categories V 9, each described by a collection of totally disconnected compact topological groups, namely the fundamental groups of the connected components of the preschemes S, S', S'', S''' . We assume for simplicity that S is

connected, and shall then give a procedure for computing its fundamental group, in terms of the fibered category formed with $\mathcal{C}'$, $\mathcal{C}''$, $\mathcal{C}'''$, suitably made explicit by means of the fundamental groups expressing these categories. The reader will note that the procedure sketched is in fact valid in the general framework of multi-Galois categories (which need not come from given preschemes S , S' , S'' , S'''). It is, moreover, the analogue of the well-known procedure for computing the fundamental group of a topological space S that is a locally finite union of closed subspaces S_i (or an arbitrary union of open subspaces S_i), by means of the fundamental groups of the components of the S_i and of the components of the $S_i \cap S_j$. Of course, the analogous situation in the framework of preschemes does fall within the general framework of descent, by introducing the prescheme S' , sum of the S_i , and the canonical morphism $g: S' \rightarrow S$.

Set

$$E' = \pi_0(S'), \quad E'' = \pi_0(S''), \quad E''' = \pi_0(S'''),$$

where π_0 denotes the functor “set of connected components”. Since the fibered products of S' over S form a simplicial object of **Sch**, it is transformed by the functor π_0 into a simplicial set of which E' , E'' , E''' are the components of dimension 0, 1, 2. We shall have to use the simplicial maps

$$q_i = \pi_0(p_i), \quad (i = 1, 2) \quad \text{and} \quad q_{ij} = \pi_0(p_{ij}), \quad (i, j) = (2, 1), (3, 2), (3, 1),$$

exhibited in the diagram

$$(1) \quad E' \xleftarrow{q_1, q_2} E'' \xleftarrow{q_{21}, q_{32}, q_{31}} E'''.$$

The objects of E' will be denoted with a prime, like s' , those of E'' resp. E''' will be denoted with a $''$ resp. a $'''$. The fact that S is connected translates into $\pi_0(K) = 0$, where K is the simplicial set defined by $g: S' \rightarrow S$, or again into the fact that the equivalence relation in E' generated by the pair of maps (q_1, q_2) is transitive.

We shall choose once and for all an element s'_0 in E' , and for every s' in E' , an element $\bar{s}' \in E''$ such that⁶

$$q_1(\bar{s}') = s'_0, \quad q_2(\bar{s}') = s',$$

thus exhibiting the connectedness of S . For every $s' \in E'$, let us choose a geometric point $\underline{s}'$ in the connected component s' of S' ; this point will in fact come in through the corresponding fiber functor $F'_{\underline{s}'}$ on the multi-Galois category $\mathcal{C}'$. The group of automorphisms of this functor, i.e. the fundamental group of S' at $\underline{s}'$, will be denoted by $\pi_{s'}$. We likewise choose points $\underline{s}''$ and $\underline{s}'''$, hence functors $F''_{\underline{s}''}$ and $F'''_{\underline{s}'''}$, whence fundamental groups $\pi_{s''}$ and $\pi_{s'''}$. Thus

$$\pi_{s'} = \pi_1(S', \underline{s}'), \quad \pi_{s''} = \pi_1(S'', \underline{s}'), \quad \pi_{s'''} = \pi_1(S''', \underline{s}''').$$

For every $s'' \in E''$, $p_1(\underline{s}'')$ lies in the same connected component as $q_1(s'')$, hence there exists an isomorphism of functors $F''_{\underline{s}''} \circ p_1^* \xrightarrow{\sim} F'_{\underline{s}'}$ (i.e. a “class of paths” from $p_1(s'')$ to $q_1(s'')$). The same remark holds for q_2 , and the q_{ij} . Let us choose all these classes of paths:

$$F''_{\underline{s}''} \circ p_i^* \xrightarrow{\sim} F'_{\underline{s}'}, \quad F'''_{\underline{s}'''} \circ p_{ij}^* \xrightarrow{\sim} F'''_{\underline{s}'''},$$

(for $i = 1, 2$ and $(i, j) = (2, 1), (3, 2), (3, 1)$). In particular there result homomorphisms of groups:

$$(2) \quad q_i^{s''} : \pi_{s''} \longrightarrow \pi_{q_i(s'')}, \quad q_{ij}^{s'''} : \pi_{s'''} \longrightarrow \pi_{q_{ij}(s''')},$$

(same values of i and (i, j)). Finally, let us recall that the structure of cloven category with fibers $\mathcal{C}'$, $\mathcal{C}''$, $\mathcal{C}'''$ also contains isomorphisms of functors:

$$p_{21}^* p_1^* \xrightarrow{\sim} p_{31}^* p_1^*, \quad p_{21}^* p_2^* \xrightarrow{\sim} p_{32}^* p_1^*, \quad p_{31}^* p_2^* \xrightarrow{\sim} p_{32}^* p_2^*,$$

⁶Note that the element $\bar{s}'$, whose existence is implicitly assumed, does not exist in all cases. Consequently, theorem 5.1 as it is stated does not apply in all cases. It is not difficult, however, taking the written text as a guide, to modify the statement of this theorem in such a way that it gives a method of computation which applies in all cases. In particular, the corollaries of the said theorem are valid as they stand.

deduced from isomorphisms of the two sides respectively with the u_i^* ($i = 1, 2, 3$), where the u_i are the three projections of S''' into S' . When one makes these data explicit, one finds, for every s''' , a well-determined element

$$(3) \quad a_i^{s'''} \in \pi_{v_i(s''')},$$

(where the v_i , $i = 1, 2, 3$ are the three maps $E''' \rightarrow E'$ defined by $v_i = \pi_0(u_i)$), subject moreover to the conditions:

$$q_1^{s''} q_{21}^{s'''} = \text{int}(a_1^{s'''} q_1^{s''} q_{31}^{s'''} \quad (s_1'' = q_{21}(s'''), s_2'' = q_{31}(s''')),$$

and the two analogous conditions, involving the a_2 and a_3 . The reader will note, moreover, that the data (1), (2), (3) make it possible to reconstruct, up to an equivalence of fibered categories, the fibered category under consideration with fibers $\mathcal{C}'$, $\mathcal{C}''$, $\mathcal{C}'''$. They must therefore make it possible in principle to reconstruct $\mathcal{C}$ up to equivalence, hence its fundamental group up to isomorphism. We shall in fact determine the fundamental group at the geometric point $p(s'_0)$ of S , i.e. the group of automorphisms of $F'_{s'_0} \circ p^*$.

We note that giving an object X' of $\mathcal{C}'$ is essentially equivalent to giving finite sets $X'_{s'}$ ($s' \in E'$) on which the $\pi_{s'}$ operate continuously. A gluing datum on such an object then amounts to giving, for every $s'' \in E''$, a bijection:

$$\varphi_{s''}: X'_{q_1(s'')} \xrightarrow{\sim} X'_{q_2(s'')}$$

compatible with the operations of $\pi_{s''}$, which operates on both sides by means of the homomorphisms $q_i^{s''}: \pi_{s''} \rightarrow \pi_{q_i(s'')}$. Taking first the s'' of the form $\overline{s'}$, we see that such a datum defines bijections

$$\psi_{s'}: X'_{s'_0} = F'_0(X') \longrightarrow X'_{s'}$$

which make it possible to identify the $X'_{s'}$ with the same set $F'_0(X') = X'_{s'_0}$, on which all the groups $\pi_{s'}$ will henceforth operate. This being so, the bijections $\varphi_{s''}$ will correspond to bijections

$$g_{s''}: F'_0(X') \xrightarrow{\sim} F'_0(X'),$$

subject on the one hand to the relations of commutation with $\pi_{s''}$:

$$a) \quad g_{s''} q_1^{s''}(g'') = q_2^{s''}(g'') g_{s''} \quad (s'' \in E'', g'' \in \pi_{s''}),$$

on the other hand to the relations

$$b) \quad \overline{g_{s'}} = \overline{g_{s'_0}} \quad (s' \in E'),$$

expressing the way in which we had identified the $X_{s'}$ with one another. When one makes explicit the condition for such a gluing datum to be in fact a descent datum, one finds the relations:

$$c) \quad a_3^{s'''} g_{q_{31}(s''')} a_1^{s'''} = g_{q_{32}(s''')} a_2^{s'''} g_{q_{21}(s''')} \quad (s''' \in E''').$$

This gives us an equivalence between the category of objects of $\mathcal{C}'$ endowed with a descent datum, and the category of finite sets on which the groups $\pi_{s'}$ operate continuously, endowed moreover with bijections $g_{s''}$ satisfying the relations a), b), c). Let then G be the group generated by the groups $\pi_{s'}$ and the new generators $g_{s''}$, subject to the relations a), b), c), and let π be the group that is the projective limit of the quotients of G by the subgroups of finite index whose inverse images in the groups $\pi_{s'}$ are open subgroups. We also say that π is the *group of Galois type generated by the $\pi_{s'}$ and the $g_{s''}$, subject to the relations a), b), c)*. One verifies at once that the category under consideration is also equivalent to the category of finite sets on which the topological group π operates continuously. This establishes the following statement:

THEOREM 5.1. *Let $g: S' \rightarrow S$ be a morphism of preschemes which is an effective descent morphism for the fibered category of étale coverings of preschemes (cf. 4.9 and 4.12). Suppose S connected, and S' , its fibered square S'' and its fibered cube S''' , sums of connected preschemes (which is the case for example if S' is of finite type over S locally noetherian and connected). Let us choose as above: a geometric point in every connected component of S' , S'' , S''' , certain classes of paths, an $s'_0 \in E'$, and for every $s' \in E'$ an $s'' \in E''$ whose two images in E' are s'_0 and s' . (E' , E'' , E''' denote respectively the set of connected components of S' , S'' , S''' .) Then the fundamental group of S at the geometric point image of s'_0 is canonically isomorphic to the group of Galois type generated by the $\pi_{s'} = \pi_1(S', \underline{s}')$ ($s' \in E'$) and generators $g_{s''}$ ($s'' \in E''$), subject to the relations a), b), c) above involving the elements of the groups $\pi_{s''} = \pi_1(S'', \underline{s}'')$, and the elements $a_i^{s'''} (i = 1, 2, 3, s''' \in E''')$ introduced above.*

COROLLARY 5.2. *Suppose that S' and S'' have only a finite number of connected components, and that the fundamental groups of the connected components of S' are topologically finitely generated. Then the fundamental group of S is topologically finitely generated.*

Thus, we shall prove later that the fundamental group of a normal projective scheme over an algebraically closed field is topologically finitely generated. Using Chow's lemma and the normalization of algebraic schemes, it will follow that the same result is true for every proper scheme over an algebraically closed field.

COROLLARY 5.3. *Suppose that S' , S'' , S''' have only a finite number of connected components, that the fundamental groups of the connected components of S' are topologically finitely presented, and the fundamental groups of the connected components of S'' topologically finitely generated. Then the fundamental group of S is topologically finitely presented.*

We note that one can express 4.9 (restricted to étale coverings) by saying that a finite radicial surjective morphism of noetherian preschemes induces an isomorphism of the fundamental groups; figuratively, one can therefore say that the fundamental group is a *topological invariant* for preschemes. More generally, one can make explicit, by means of 5.1, the effect on the fundamental group of operations on preschemes having a simple topological meaning, such as the “pinching” of the prescheme along a finite set of points. One finds for example:

COROLLARY 5.4. *Let $g: S' \rightarrow S$ be a finite morphism of finite presentation, T a discrete subset of S . For every $s \in S$, let $n(s)$ be the “geometric number of points” in the fiber $g^{-1}(s)$ (which can also be expressed as the separable degree of $g^{-1}(s)$ over $k(s)$, the sum of the separable degrees of its residue extensions). Suppose that for $s \in S - T$ we have $n(s) = 1$. For every $s \in T$, let K_s be an algebraically closed extension of $k(s)$, I_s the set of geometric points of S' with values in K_s (it is a set with $n(s)$ elements), I'_s the complement of a chosen point of I_s , and finally I' the set that is the union of the I'_s . Suppose S' connected. Then the fundamental group of S is isomorphic to the group of Galois type generated by the fundamental group of S' and generators $g_i (i \in I')$, subject to no additional condition.*

The details of the proof are left to the reader; the statement obtained is merely the translation, into the language of group theory, of the fact that we have an equivalence between the category C of étale coverings of S and the category of étale coverings X' of S' endowed, for every $s \in T$, with a transitive system of bijections between the $n(s)$ fibers of X' at the points of $g^{-1}(s)$ with values in K_s . (In this intrinsic form, of course, it is no longer necessary to assume S' connected.)

EXAMPLE 5.5. One easily proves that the rational curve $\mathbf{P}_k^1$ over an algebraically closed field k is simply connected⁷. Hence the fundamental group of a complete rational curve having exactly one double point, with n analytic branches, is the free group of Galois type generated by $n - 1$ generators. For example, in the case of an ordinary double point, one finds the fundamental group $\widehat{\mathbf{Z}}$, as announced in (I 11 a)). On the other hand, the existence

⁷Cf. Exp. XI 1.1.

of a cusp (which is a “geometrically unibranch” point) has no influence on the fundamental group.

COROLLARY 5.6. *Let $g: S' \rightarrow S$ be a universally submersive morphism of preschemes with geometrically connected fibers, S being connected. Then S' is connected, and choosing a geometric point s' in S' and denoting by s its image in S , the homomorphism*

$$\pi_1(S', s') \longrightarrow \pi_1(S, s)$$

is surjective. If g is an effective descent morphism for the fibered category of étale coverings of preschemes (cf. 4.12), then, introducing the geometric point $s'' = \text{diag}(s')$ of $S'' = S' \times_S S'$, and the two homomorphisms

$$p_{1*}, p_{2*}: \pi_1(S'', s'') \longrightarrow \pi_1(S', s')$$

induced by the two projections, $\pi_1(S, s)$ is isomorphic to the cokernel of this pair of morphisms in the category of groups of Galois type, i.e. to the quotient of $\pi_1(S', s')$ by the closed invariant subgroup generated by the elements of the form $p_{1}(g'')p_{2*}(g'')^{-1}$, with $g'' \in \pi_1(S'', s'')$.*

Indeed, we know by 3.4 that the functor $X \mapsto X \times_S S'$ from étale coverings over S to étale coverings over S' is fully faithful, which is equivalent to the fact that the homomorphism on the fundamental groups is an epimorphism (V 6.9). The last assertion is an immediate consequence of the description 5.1.

REMARK 5.7. It is not known at present whether the fundamental group of a proper scheme over an algebraically closed field k is topologically finitely presented⁸. Using 5.3, a well-known technique of hyperplane sections, and the desingularization of normal surfaces, one is reduced to the case of a *smooth surface over k* . This makes it possible at least to show, by transcendental means, that the answer is affirmative in characteristic 0 (and this without being obliged to assume the triangulability of singular algebraic varieties). In characteristic $p > 0$, the main difficulty seems to lie in the case of curves, of which one knows only that the fundamental group is a quotient of the one that occurs in the classical case (cf. the following exposé), the kernel by which one divides being, however, very poorly known.

REMARK 5.8. One could make explicit other special cases than 5.4 and 5.6 where 5.1 takes a particularly simple form. An interesting case is the one where S is the quotient of S' by a finite group of automorphisms Γ . Then the category of étale coverings of S' is equivalent to the category of étale coverings X' of S' on which the group Γ operates compatibly with its operations on S' , in such a way that for every $s' \in S'$ and every $g \in \Gamma_{s'}$ (where $\Gamma_{s'}$ denotes the inertia group of s' in Γ), g operates trivially on the fiber $X'_{s'}$. If S' is connected, this statement can be interpreted as follows. Let C'_0 be the category of étale coverings of S' on which Γ operates compatibly with its operations on S' (but without necessarily satisfying the above condition on the inertia groups of the points of S'). One easily sees that it is a Galois category (V 5), and that for every geometric point a' of S' , the fiber functor $X' \mapsto X'_{a'}$ on C'_0 is a fundamental functor. Let $\pi_1(S', \Gamma; a') = G$ be the group of automorphisms of this functor, endowed with its usual topology. We then have a canonical exact sequence

$$e \longrightarrow \pi_1(S', a') \longrightarrow G \longrightarrow \Gamma \longrightarrow e$$

(special case of (V 6.13), where one takes for S the trivial covering $S' \times \Gamma$ of S' defined by Γ , where one lets Γ operate in the obvious way). Moreover, for every geometric point b' of S' , we have an isomorphism $\pi_1(S', \Gamma; b') \rightarrow G = \pi_1(S', \Gamma; a')$ defined up to an inner automorphism coming from $\pi_1(S', a')$, and since $\Gamma_{b'}$ maps in an obvious way into the left-hand side, we obtain a homomorphism

$$u_{b'}: \Gamma_{b'} \longrightarrow G,$$

⁸This seems very improbable in the case of smooth curves of genus $g \geq 2$, in characteristic $p > 0$. When one replaces π_1 by its largest quotient prime to p , on the other hand, it seems that the well-known techniques make it possible to give an affirmative answer, even without a properness hypothesis. Cf. a work in preparation by J.P. Murre.

defined up to an inner automorphism (coming from $\pi_1(S', a')$), whose composite with the canonical homomorphism $G \rightarrow \Gamma$ is, moreover, the canonical immersion $\Gamma_{b'} \rightarrow \Gamma$. This being so, *the fundamental group $\pi_1(S, a)$ is canonically isomorphic to the quotient group of $G = \pi_1(S', \Gamma; a')$ by the closed invariant subgroup generated by the images of the homomorphisms $\Gamma_{b'} \rightarrow G$* . In particular, the image of $\pi_1(S', a')$ in $\pi_1(S, a)$ is an invariant subgroup, and the corresponding quotient is isomorphic to a quotient of Γ . One can, moreover, reduce the number of “relations” introduced by introducing, for every $g \in \Gamma$, $g \neq e$, the subscheme S'_g of coincidences of the automorphisms id_S and g of S , by choosing a geometric point $b'_{g,i}$ in each connected component of S'_g , and then one of the corresponding homomorphisms $\pi_1(S', \Gamma; b'_{g,i}) \rightarrow G$, whence liftings $\bar{g}_i$ of g in G . It then suffices to take the quotient of G by the closed invariant subgroup of G generated by the $\bar{g}_i$.

When a' is invariant under Γ , one easily sees that Γ operates in a natural way on $\pi_1(S', a')$, and G is identified with the corresponding semidirect product. Then identifying Γ with a subgroup of G , one sees that in the relations introduced above, taking $b' = a'$, one finds “ $g = e$ ” for $g \in I$. Hence *if S' has a geometric point a' fixed by I (i.e. a point s' whose inertia group is Γ), then $\pi_1(S, a)$ is a quotient group of the quotient group of Galois type of $\pi_1(S', a')$ obtained by “making trivial” the operations of Γ on $\pi_1(S', a')$; and it is even isomorphic to this latter group if one assumes that for every $g \in G$, the inertia set S'_g is connected, hence passes through the locality of a'* . This last assertion is indeed contained in the second description given above for the relations to be introduced in G .

This last result applies in particular if one takes for S' the cartesian power X^n of a connected prescheme over an algebraically closed field, for Γ the symmetric group $\Gamma = \mathfrak{S}_n$, operating in the usual way, whence for S the n th symmetric power of S . Taking then for a' a geometric point localized at the diagonal, we are under the preceding conditions, the inertia sets S'_g indeed all containing the diagonal. Using the fact, proved in the following exposé, that if X is proper and connected over k , the fundamental group of X^n is identified with $\pi_1(X)^n$, we find the following amusing result: *If X is proper and connected over k algebraically closed, the fundamental group of its n th symmetric power, $n \geq 2$, is isomorphic to the fundamental group of X made abelian*. (I do not know whether the analogous fact in algebraic Topology is known; it should be possible to establish it by the same method of descent.) Taking for example for X a rational curve $X = \mathbf{P}_k^1$, we find an N th proof of the fact that $\mathbf{P}_k^r$ is simply connected, using the fact that $\mathbf{P}_k^1$ is. Let us now take for X a simple curve over k , and $n \geq 2g - 1$, so that $\text{Sym}^n(X)$ is fibered over the jacobian J , with projective spaces as fibers, hence (as we shall see by means of the results of the two following exposés) has the same fundamental group as J . We then recover without dévissage the well-known fact that *the fundamental group of the jacobian of X is isomorphic to the fundamental group of X made abelian*.

6. A fundamental exact sequence. Descent by morphisms with relatively connected fibers

THEOREM 6.1. *Let S be the spectrum of an artinian ring A with residue field k , $\bar{k}$ an algebraic closure of k , X an S -prescheme, $X_0 = X \otimes_A k$, $\bar{X}_0 = X \otimes_A \bar{k}$, $\bar{a}$ a geometric point of $\bar{X}$, a its image in X , b its image in S . Suppose that X_0 is quasi-compact and geometrically connected over k (N.B. if X is proper over S , this means that $H^0(X_0, \mathcal{O}_{X_0})$ is a local artinian ring with residue field radicial over k). Then the sequence of canonical homomorphisms*

$$e \longmapsto \pi_1(\bar{X}_0, \bar{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(S, b) \longrightarrow e$$

is exact, and one has

$$\pi_1(S, b) \stackrel{\sim}{\longleftarrow} \pi_1(k, \bar{k}) = \text{Galois group of } \bar{k} \text{ over } k.$$

Since fundamental groups do not change on dividing out by the nilpotent elements, one may assume $A = k$, which already makes the last isomorphism evident. Let k' be the separable

closure of k in $\bar{k}$, and consider $X' = X \otimes_k k'$, and the image a' of $\bar{a}$ in X' . One has a sequence of canonical homomorphisms

$$e \mapsto \pi_1(\bar{X}_0, \bar{a}) \longrightarrow \pi_1(X', a') \longrightarrow \pi_1(S', b') \longrightarrow e$$

(where $S' = \text{Spec}(k')$). Finally, one has a canonical homomorphism from this sequence into the one relative to X/k , thanks to the diagram

$$\begin{array}{ccccc} S & \longleftarrow & X & \longleftarrow & \bar{X}_0 \\ \uparrow & & \uparrow & & \parallel \\ S' & \longleftarrow & X' & \longleftarrow & \bar{X}_0 \end{array}$$

On the other hand, one sees that this homomorphism of sequences of groups is an isomorphism, as follows from 4.11. One is therefore reduced to proving that the second sequence is exact, i.e. one may assume that k is *perfect*. Let then k_i be the finite Galois subextensions of k in $\bar{k}$, set $X_i = X \otimes_k k_i$, and let a_i be the image of $\bar{a}$ in X_i . We leave to the reader the task of verifying that the natural homomorphism

$$\pi_1(\bar{X}_0, a) \xrightarrow{\sim} \varprojlim \pi_1(X_i, a_i)$$

is an isomorphism, which simply means that an étale covering of $\bar{X}$ comes from an étale covering of some X_i , and that the latter is essentially unique, up to passing to some X_j , $j \geq i$. On the other hand, let π_i be the Galois group of k_i over k , i.e. the group opposite to the group of S -automorphisms of $S_i = \text{Spec}(k_i)$. Since the functor $S' \mapsto X \times_S S'$ from the étale coverings of S to the étale coverings of X is fully faithful (3.4), it follows that π_i is also isomorphic to the group opposite to the groups of X -automorphisms of the connected principal covering X_i of X . It therefore follows from V 6.13 that one has an exact sequence

$$e \mapsto \pi_1(X_i, a_i) \longrightarrow \pi_1(X, a) \longrightarrow \pi_i \longrightarrow e.$$

Passing to the projective limit over i in these exact sequences, one finds an exact sequence (since one is in the category of groups of Galois type), which is none other than the sequence considered in 6.1. This completes the proof.

The translation of the right exactness in 6.1 into geometric language is the following:

COROLLARY 6.2. *With the preceding notation, let X' be an étale covering of X , and let $\bar{X}'_0$ be the corresponding étale covering of $\bar{X}_0$. The following conditions are equivalent:*

- (i) *There exists an S' étale over S , and an X -isomorphism $X' \xrightarrow{\sim} X \times_S S'$, (S' is then determined up to unique isomorphism by virtue of 3.4).*
- (ii) *$\bar{X}'_0$ is completely decomposed over $\bar{X}_0$.*
If X' is connected, these conditions are also equivalent to:
- (ii bis) *$\bar{X}'_0$ has a section over $\bar{X}_0$.*

(N.B. This last complement is essential; the equivalence of (i) and (ii) means only that $\pi_1(S, b)$ is the quotient group of $\pi_1(X, a)$ by the closed invariant subgroup generated by the image of $\pi_1(\bar{X}_0, \bar{a})$, and not by this image itself). Under the preceding conditions, we shall say that X' is a *geometrically trivial* covering of X .

REMARK 6.3. In the statement 6.1 one cannot replace $\bar{k}$ by an arbitrary algebraically closed extension of k , even if k is already assumed algebraically closed. In other words, it is not true in general that if X is a connected algebraic scheme over an algebraically closed field k , its fundamental group does not change on replacing k by an algebraically closed extension; this is already false for example in characteristic $p > 0$ for the affine line over k , because of the phenomena of “higher ramification” at the point at infinity, implying a “continuous” structure for the fundamental group. We shall see however in the next exposé that such phenomena cannot occur if X is *proper* over k . We shall also show by transcendental means that the same holds if k is of characteristic zero.

COROLLARY 6.4. *Suppose that a is localized at an $x \in X$ which is rational over k (or more generally, which has a residue field radicial over k). Then the exact sequence 6.1 is split.*

One may assume $S = \text{Spec}(k)$. If x is rational over k , it corresponds to a section $S \rightarrow X$ of X over S , transforming b into a , and defining a homomorphism $\pi_1(S, b) \rightarrow \pi_1(X, a)$ which is the desired splitting. If $k(x)$ is radicial over k , one reduces to the preceding case by making the base extension $\text{Spec}(k(x)) \rightarrow \text{Spec}(k)$.

THEOREM 6.5. *Let $f: X \rightarrow S$ be a proper and surjective morphism of finite presentation, with geometrically connected fibers, X' a prescheme of finite presentation and proper over X , s a point of S , $F = X_s$ the fiber of X at s , and F'_1 a connected component of the fiber $F' = X'_s$ of X' at s . In order that there exist an open neighborhood X'_1 of F'_1 in X' , an étale S -scheme S'_1 and an X -isomorphism $X'_1 \xrightarrow{\sim} S'_1 \times_S X$, it is necessary and sufficient that X' be étale over X at the points of F'_1 , and that F'_1 be a geometrically trivial covering of F .*

The necessity of the condition being trivial, it remains to prove the sufficiency. One easily reduces to the case where S is noetherian. Consider the Stein factorization $X \rightarrow T \rightarrow S$ of f , where T is the spectrum of the Algebra $f_*(\mathcal{O}_X)$ over S . Since the fibers of X over S are geometrically connected, and f is surjective, the morphism $T \rightarrow S$ is finite surjective and radicial, hence (4.10) every T' étale over T comes by inverse image from an S' étale over S . This reduces us to proving 6.5 with S replaced in it by T , i.e. to the case where one assumes $f_*(\mathcal{O}_X) = \mathcal{O}_S$. Consider then the Stein factorization $X' \rightarrow S' \rightarrow S$ of the proper morphism $h: X' \rightarrow S$, where S' is the spectrum of the Algebra $h_*(\mathcal{O}_{X'})$. The morphisms $X' \rightarrow X$ and $X' \rightarrow S'$ define a canonical morphism

$$X' \longrightarrow X \times_S S',$$

and our assertion is contained in the following one:

COROLLARY 6.6. *Let $f: X \rightarrow S$ be a proper morphism of locally noetherian preschemes, such that $f_*(\mathcal{O}_X) = \mathcal{O}_S$, and let X' be a prescheme proper over X . Consider the Stein factorization $X' \rightarrow S' \rightarrow S$ for $X' \rightarrow S$, and the canonical morphism $X' \rightarrow X \times_S S'$. Let s be a point of S , s' a point of S' over s , corresponding to a connected component F'_1 of the fiber X'_s of X' at s . In order that the morphism $X' \rightarrow X \times_S S'$ be an isomorphism over an open neighborhood U' of s' étale over S , it is necessary and sufficient that X' be étale over X at the points of F'_1 , and that F'_1 be a geometrically trivial covering of the fiber $F = X_s$.*

The necessity again being trivial, it remains to prove the sufficiency. The conclusion also means that a) the morphism deduced from $X' \rightarrow X \times_S S'$ by the change of base $\text{Spec}(\widehat{\mathcal{O}}_{s'}) \rightarrow S'$ is an isomorphism, b) S' is étale over S at s' , i.e. $\widehat{\mathcal{O}}_{s'}$ is étale over $\widehat{\mathcal{O}}_s$. In this form, one sees that the conclusion is invariant under the change of base $\text{Spec}(\widehat{\mathcal{O}}_s) \rightarrow S$. The hypotheses being likewise stable under this change of base, one may therefore assume that S is the spectrum of a complete noetherian local ring. One may moreover obviously assume X' connected, which implies here that $S' = \text{Spec}(\mathcal{O}_{s'})$, $F' = F'_1$.

Since the set of points of X' where X' is étale over X is open and contains the fiber $X'_{s'} = F'$, X' being proper over S , it follows that X' is étale over X . Since it induces on $F = X_s$ an étale covering isomorphic to some $F \otimes_{k(s)} L$, where L is étale over $k(s)$, it follows from 1.10 that it is isomorphic to a covering of the form $X \times_S T$, with T étale over S . (N.B. here again, it suffices to use that the functor of 1.10 is fully faithful, which follows from the fact that a formal isomorphism of coherent sheaves on X comes from an isomorphism of these sheaves). Hence, if T is defined by the algebra B finite over A , X' is identified with the spectrum of the Algebra $\mathcal{O}_X \otimes_A B$ over X , from which it follows at once, since $f_*(\mathcal{O}_X) = \mathcal{O}_S$, that $h_*(\mathcal{O}_{X'})$ is defined by B , hence the canonical homomorphism $X' \rightarrow X \times_S S'$ is none other than the isomorphism under consideration $X' \xrightarrow{\sim} X \times_S T$. This completes the proof.

COROLLARY 6.7. *Under the conditions of 6.5, in order that there exist a prescheme S' étale over S and an X -isomorphism $X' \xrightarrow{\sim} X \times_S S'$, it is necessary and sufficient that X' be étale over X and that for every $s \in S$, X'_s be a geometrically trivial covering of X_s .*

Indeed, if this is so, X' is the union of open sets X'_i which are isomorphic to inverse images of S'_i étale over S . One then easily sees that these S'_i glue into an S' étale over S , and that one obtains an isomorphism $X' \xrightarrow{\sim} X \times_S S'$. One can say for example that the X'_i are endowed with descent data relative to $X \rightarrow S$, which necessarily glue into a descent datum on the whole of X' relative to $X \rightarrow S$. And since the latter is effective on the X'_i , it follows easily (thanks to a sorites forgotten in no. 4) that it is effective. One can also state 6.7 in the following form:

COROLLARY 6.8. *Let $f: X \rightarrow S$ be a proper surjective morphism of finite presentation, with geometrically connected fibers. Then f is an effective descent morphism for the fibered category of preschemes finite étale over others. The functor $S' \mapsto X \times_S S'$ induces an equivalence of the category of preschemes étale and finite over S with the category of preschemes étale and finite over X which induce on each fiber X_s a geometrically trivial covering.*

REMARK 6.9. Let $f: X \rightarrow S$ be a proper and surjective morphism, with Y locally noetherian; then f factors into a morphism $X \rightarrow S'$ satisfying the hypothesis of 6.8, and a finite surjective morphism $S' \rightarrow S$ to which 4.7 applies, hence into a product of two morphisms which are *universal effective descent morphisms* for the fibered category of preschemes étale and finite over others. One can conclude from this that f itself is a universal effective descent morphism for the fibered category under consideration. One thus recovers 4.12 by a different method.

REMARK 6.10. The conclusion of 6.7 does not remain valid if one replaces the hypothesis that f is proper by: X is of finite type over S and admits a section over S (so that f is universally submersive and a descent morphism for the fibered category of preschemes étale over others), even when S is the spectrum of a discrete valuation ring and when X' is an étale covering of X . To see this, one starts from a Z proper over S , whose general fiber is a nonsingular rational curve, and whose special fiber Z_0 consists of two concurrent lines. For example, if t is a uniformizer of the valuation ring A , one takes the closed subscheme Z of $\mathbf{P}_A^2$ defined by the homogeneous equation $x^2 + y^2 + tz^2 = 0$ (homogeneous coordinates x, y, z). One takes for X the complement of the singular point a of Z_0 in the union $Z \cup \mathbf{P}_k^2$. The fibers of X are $\mathbf{P}_k^1$ and $\mathbf{P}_k^2 - a$, hence are geometrically simply connected (i.e. every étale covering of such a fiber is geometrically trivial). However one easily constructs, proceeding as in no. 4, étale coverings of X which do not come from étale coverings of S , by gluing trivial coverings of $Z - a$ and of $\mathbf{P}_k^2 - a$. It is possible on the other hand that the conclusion of 6.7 persists if one replaces in it the hypothesis of properness by that X be *universally open* of finite presentation over S^9 . This is true at least if one assumes that the fibers of X over S are geometrically irreducible, and not only geometrically connected. Let us only point out that in this question one can reduce to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field.

The interpretation of 6.7 in terms of the fundamental group is the following:

COROLLARY 6.11. *Let $f: X \rightarrow S$ be a proper surjective morphism of finite presentation, with geometrically connected fibers. Suppose X , hence S , connected. Let a be a geometric point of X , b its image in S , and for every $s \in S$, let us choose an algebraic closure $\bar{k}(s)$ of $k(s)$, a geometric point a_s of S_s with values in this extension, and a class of paths from a_s to a , whence a homomorphism $\pi_1(\bar{X}_s, a_s) \rightarrow \pi_1(X, a)$, where $\bar{X}_s = X_s \otimes_{k(s)} \bar{k}(s)$. Then the homomorphism $\pi_1(X, a) \rightarrow \pi_1(S, b)$ is surjective, and its kernel is the closed invariant subgroup of $\pi_1(X, a)$ generated by the images of the $\pi_1(\bar{X}_s, a_s)$.*

⁹This is now proved, g being only universally open and surjective; cf. SGA 4 XV 1.15.

REMARK 6.12. Under the conditions of 6.7, assuming S noetherian, one easily sees that the set of points $s \in S$ such that $S'_{s'}$ is geometrically trivial over X_s is a constructible set; if X' is proper over X , it is even open, as one sees from 6.6. This therefore allows one, if S is a Jacobson prescheme (for example of finite type over a field), or X' is proper over X , to restrict oneself, in order to verify the conditions of 6.7, to the points s of S which are *closed*. Likewise, in 6.11 it then suffices to take the $\pi_1(\overline{X}_s, a_s)$ for the points of S which are closed.

7. Bibliography

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